ONE HOT ENCODING

One hot encoding is a process by which categorical variables are converted into a form that could be provided to ML algorithms to do a better job in prediction.

We can do this by first encoding the strings consistently to integers using the scikit-learn class LabelEncoder. Then convert the vector of integers to a one hot encoding using the Keras function to_categorical( ).

Label Encoding

Food Name	Categorical #	Calories
Apple	1	95
Chicken	2	231
Broccoli	3	50

→

One Hot Encoding

Apple	Chicken	Broccoli	Calories
1	0	0	95
0	1	0	231
0	0	1	50

```
from numpy import array
from sklearn.preprocessing import LabelEncoder


data = ['Amman', 'Salt', 'Zarqa', 'Amman', 'Mada
ba','Salt','Zarqa','Amman','Amman']
values = array(data)
print(values)

# integer encode
label_encoder = LabelEncoder()
integer_encoded = label_encoder.fit_transform(values)
print(integer_encoded)

from tensorflow.keras.utils import to_categorical
# one hot encode
encoded = to_categorical(integer_encoded)
print(encoded)
```

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder

df = pd.read_csv("data.csv",  delimiter=';')
print (df.shape[0])
array = df.values
values = array[:, 1]
print (values)

# integer encode
label_encoder = LabelEncoder()
integer_encoded = label_encoder.fit_transform(values)
print(integer_encoded)

from tensorflow.keras.utils import to_categorical
# one hot encode
encoded = to_categorical(integer_encoded)
print(encoded)

import numpy as np
new_array = np.concatenate([array[:,0], array[:,2]])
new_array = new_array.reshape(2,df.shape[0])
new_array = new_array.transpose()
print (new_array)

X = np.concatenate((new_array,encoded),axis=1)
print (X)

Y = array[:, 3]
```